Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

RealMLP XAI study notebook

This notebook is a model-specific XAI study for the saved 10-fold RealMLP checkpoint family reconstructed from `churn_analysis_helpers.py`.

The focus is not merely to replay predictions. The focus is to understand the model's smooth nonlinear response geometry, feature sensitivity, and where that geometry helps or fails.

Study perspective

We treat the fitted predictor as an ensemble-averaged neural probability map:

$$p_{\mathrm{mlp}}(x) = \frac{1}{K}\sum_{k=1}^{K} \mathrm{softmax}\big(g_k(\phi_{\mathrm{num}}(x_{\mathrm{num}}), \phi_{\mathrm{cat}}(x_{\mathrm{cat}}))\big)_1.$$

Unlike the tree model, the RealMLP combines:

- robust smooth-clipped numerical scaling,
- periodic numerical embeddings,
- one-hot and learnt categorical embeddings,
- stacked GELU nonlinearities,
- fold-model averaging.

The central XAI questions are therefore:

1. Which variables drive the ensemble's global behaviour?
2. Where is the response surface smooth, sharp, or unstable?
3. Which cases create member disagreement?
4. Which inputs move the model most under local perturbation or ablation?

Required files in `./`

- `train.csv`
- `test.csv`
- `realmlp_fold_*.pth`

- `churn_analysis_helpers.py`

In [2]:
```
from pathlib import Path
import json
import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import torch
import shap

from sklearn.metrics import roc_auc_score, log_loss, brier_score_lo
from sklearn.inspection import permutation_importance

from churn_analysis_helpers import (
    CFG,
    seed_everything,
    setup_plotting,
    ensure_output_dir,
    savefig,
    build_feature_frames,
    prepare_realmlp_frames,
    inspect_realmlp_schema,
    replay_realmlp,
    make_decile_frame,
    cumulative_capture,
    FE_realmlp,
    RobustScaleSmoothClipTransform,
    RealMLP,
    load_realmlp_checkpoints,
)
from sklearn.preprocessing import TargetEncoder

seed_everything(67)
setup_plotting()

cfg = CFG()
ensure_output_dir(cfg)

# Optional path overrides
# cfg.train_path = "./train.csv"
# cfg.test_path = "./test.csv"
# cfg.realmlp_pattern = "realmlp_fold_*.pth"
# cfg.output_dir = "./analysis_outputs_realmlp"
```

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1. Reconstruct the exact RealMLP feature space

This section uses the helper file exactly as written to rebuild the numerical and categorical representations used by the saved checkpoints.

In [3]:
```
bundle = build_feature_frames(cfg)
realmlp_bundle = prepare_realmlp_frames(bundle)
schema = inspect_realmlp_schema(realmlp_bundle, cfg)

print("Numerical features:", schema["numeric_count"])
print("Categorical features:", schema["categorical_count"])
print("Target-encoded cats:", schema["target_encoded_cat_count"])
print("Augmented numeric dim:", schema["augmented_numeric_count"])
display(pd.Series(schema).to_frame("value"))
```

```
Numerical features: 22
Categorical features: 54
Target-encoded cats: 32
Augmented numeric dim: 54
```

	value
numeric_count	22
categorical_count	54
target_encoded_cat_count	32
augmented_numeric_count	54
NUMS	[SeniorCitizen, tenure, MonthlyCharges, TotalC...
CATEGORY	[gender, Partner, Dependents, PhoneService, Mu...

Mathematical framing

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2. Replay the saved fold checkpoints

The helper already reconstructs the exact replay path. We use that first, then build additional XAI routines around the replayed pipeline.

In [8]:
```
replay = replay_realmlp(bundle, realmlp_bundle, cfg)

print("device:", replay["device"])
print("RealMLP OOF AUC:", f"{replay['realmlp_oof_auc']:.6f}")
print("RealMLP OOF log loss:", f"{log_loss(bundle['y'], replay['rea
print("RealMLP OOF Brier:", f"{brier_score_loss(bundle['y'], replay
```

```
device: cpu
RealMLP OOF AUC: 0.917199
RealMLP OOF log loss: 0.298696
RealMLP OOF Brier: 0.095583
```

In [9]:

```
analysis_train = pd.DataFrame({
    cfg.id_col: bundle["train"][cfg.id_col].values,
    "y_true": bundle["y"],
    "realmlp_pred": replay["realmlp_oof"],
})

analysis_test = pd.DataFrame({
    cfg.id_col: bundle["test"][cfg.id_col].values,
    "realmlp_pred": replay["realmlp_test_mean"],
})

analysis_train["pred_rank_pct"] = analysis_train["realmlp_pred"].ra
analysis_train["hard_label"] = (analysis_train["realmlp_pred"] >= 0
analysis_train["error"] = analysis_train["y_true"] - analysis_train
analysis_train["abs_error"] = analysis_train["error"].abs()
analysis_train["case_type"] = np.select(
    [
        (analysis_train["y_true"] == 1) & (analysis_train["hard_lab
        (analysis_train["y_true"] == 0) & (analysis_train["hard_lab
        (analysis_train["y_true"] == 0) & (analysis_train["hard_lab
        (analysis_train["y_true"] == 1) & (analysis_train["hard_lab
    ],
    ["TP", "TN", "FP", "FN"],
    default="other"
)
analysis_train["risk_band"] = pd.qcut(
    analysis_train["realmlp_pred"], q=5,
    labels=["very_low", "low", "mid", "high", "very_high"],
    duplicates="drop",
)

member_cols = [f"member_{i}" for i in range(1, replay["realmlp_test_
test_member_df = pd.DataFrame(replay["realmlp_test_members"], colum
analysis_test = pd.concat([analysis_test, test_member_df], axis=1)
analysis_test["member_std"] = analysis_test[member_cols].std(axis=1
analysis_test["member_range"] = analysis_test[member_cols].max(axis

analysis_train.head()
```

Out[9]:

	id	y_true	realmlp_pred	pred_rank_pct	hard_label	error	abs_erro
0	0	0	0.009655	0.282337	0	-0.009655	0.00965
1	1	0	0.002292	0.044788	0	-0.002292	0.00229
2	2	0	0.301171	0.704309	0	-0.301171	0.30117
3	3	1	0.707725	0.927854	1	0.292275	0.29227
4	4	1	0.788549	0.960841	1	0.211451	0.21145

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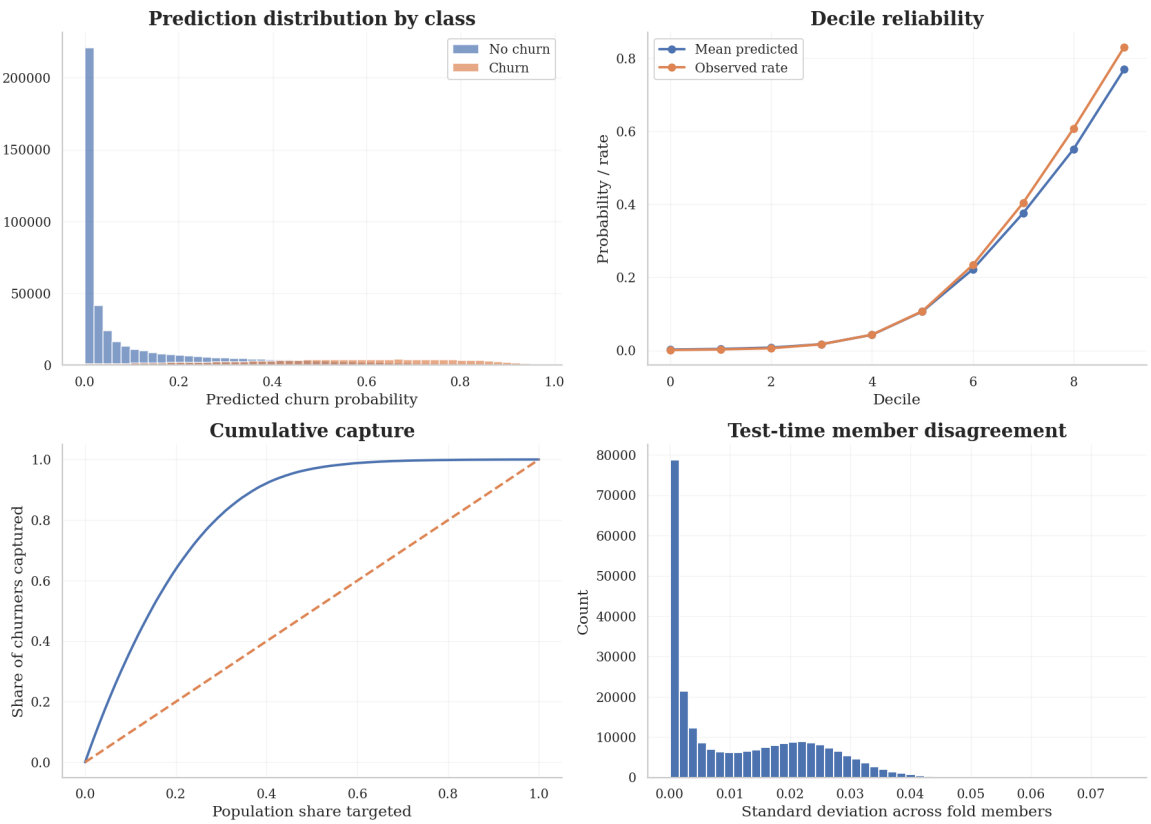
3. Behavioural diagnostics

These plots answer whether the replayed ensemble:

- separates classes cleanly,
- calibrates sensibly,
- concentrates positives in the upper tail,
- produces tail overconfidence or underconfidence.

In [10]:
```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

axes[0, 0].hist(analysis_train.loc[analysis_train["y_true"] == 0, "
axes[0, 0].hist(analysis_train.loc[analysis_train["y_true"] == 1, "
axes[0, 0].set_title("Prediction distribution by class")
axes[0, 0].set_xlabel("Predicted churn probability")
axes[0, 0].legend()

dec_realmlp = make_decile_frame(bundle["y"], analysis_train["realml
axes[0, 1].plot(dec_realmlp["mean_pred"], marker="o", label="Mean p
axes[0, 1].plot(dec_realmlp["churn_rate"], marker="o", label="Obser
axes[0, 1].set_title("Decile reliability")
axes[0, 1].set_xlabel("Decile")
axes[0, 1].set_ylabel("Probability / rate")
axes[0, 1].legend()

cap_realmlp = cumulative_capture(bundle["y"], analysis_train["realm
axes[1, 0].plot(cap_realmlp["population_share"], cap_realmlp["captu
axes[1, 0].plot([0, 1], [0, 1], linestyle="--")
axes[1, 0].set_title("Cumulative capture")
axes[1, 0].set_xlabel("Population share targeted")
axes[1, 0].set_ylabel("Share of churners captured")

axes[1, 1].hist(analysis_test["member_std"], bins=50)
axes[1, 1].set_title("Test-time member disagreement")
axes[1, 1].set_xlabel("Standard deviation across fold members")
axes[1, 1].set_ylabel("Count")

plt.tight_layout()
savefig(cfg, "realmlp_01_core_diagnostics.png")
plt.show()

display(dec_realmlp)
```

```
saved: analysis_outputs\realmlp_01_core_diagnostics.png
```



	bin	mean_pred	churn_rate	n	pred_min	pred_max
0	0	0.002330	0.001094	59420	0.000880	0.003107
1	1	0.004047	0.002710	59419	0.003107	0.005285
2	2	0.007776	0.005840	59419	0.005285	0.011012
3	3	0.017026	0.016829	59420	0.011013	0.025836
4	4	0.042658	0.042966	59419	0.025837	0.065962
5	5	0.106499	0.107390	59419	0.065962	0.156202
6	6	0.221997	0.233793	59420	0.156208	0.294355
7	7	0.375918	0.404180	59419	0.294356	0.460177
8	8	0.551160	0.607600	59419	0.460180	0.647994
9	9	0.768880	0.829670	59420	0.648000	0.966818

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4. Build reusable prediction functions for custom slices

The helper already provides replay for the original train/test split. For XAI work we also need predictions on modified input frames.

The next cell wraps the same preprocessing logic used by the helper so that we can:

- perturb rows,
- draw two-dimensional conditional slices,
- compute finite-difference sensitivities,
- run grouped ablations.

This cell does not modify the helper file. It only uses it as-is.

In [11]:
```
def prepare_custom_realmlp_df(raw_df, realmlp_bundle, cfg):
    df = raw_df.copy()
    if cfg.id_col in df.columns:
        df = df.drop(columns=[cfg.id_col]).copy()

    if cfg.target_col in df.columns:
        df[cfg.target_col] = (df[cfg.target_col] == "Yes").astype(i

    df = FE_realmlp(df, cfg)

    for c in ["MonthlyCharges", "TotalCharges"]:
        for n_bins in ([100, 1000] if c == "MonthlyCharges" else [5
            kb = realmlp_bundle["kb_objects"][f"{c}_bin{n_bins}"]
            vals = df[c].values.reshape(-1, 1)
            df[f"{c}_bin{n_bins}"] = kb.transform(vals).flatten().a

    drop_cols = [c for c in realmlp_bundle["DROP"] if c in df.colum
    df = df.drop(columns=drop_cols, errors="ignore")

    if "service_count" not in df.columns:
        from churn_analysis_helpers import REALMLP_SERVICE_COLS
        df["service_count"] = (df[REALMLP_SERVICE_COLS] == "Yes").s

    category = realmlp_bundle["CATEGORY"]
    nums = realmlp_bundle["NUMS"]

    for c in category:
        if c not in df.columns:
            df[c] = "__missing__"
        mapping_info = realmlp_bundle["category_mappings"][c]
        mapping = mapping_info["mapping"]
        default = mapping_info["default"]
        df[c] = df[c].map(lambda x: mapping.get(x, default)).astype

    for c in nums:
        if c not in df.columns:
            df[c] = 0.0

    X_num = df[nums].values.astype(np.float32)
    X_cat = df[category].values.astype(np.int64)
```

```
    return df, X_num, X_cat


def build_realmlp_fold_cache(realmlp_bundle, cfg, embed_dim=10, n_e
    device = torch.device("cuda" if torch.cuda.is_available() else
    X_num = realmlp_bundle["X_num"]
    X_cat = realmlp_bundle["X_cat"]
    y_nn = realmlp_bundle["y_nn"]
    cat_dims_global = realmlp_bundle["cat_dims_global"]
    cols_to_encode = realmlp_bundle["cols_to_encode"]
    ckpts = load_realmlp_checkpoints(cfg)

    from sklearn.model_selection import StratifiedKFold
    skf = StratifiedKFold(n_splits=cfg.n_splits, shuffle=True, rand

    cache = []
    for fold, ((train_idx, valid_idx), ckpt_path) in enumerate(zip(
        X_num_train, X_cat_train, y_train = X_num[train_idx], X_cat

        te = TargetEncoder(target_type="binary", cv=5)
        if len(cols_to_encode) > 0:
            X_cat_train_encoded = te.fit_transform(X_cat_train[:, c
        else:
            X_cat_train_encoded = np.zeros((len(train_idx), 0), dty

        X_num_train_aug = np.concatenate([X_num_train, X_cat_train_
        rssc = RobustScaleSmoothClipTransform().fit(X_num_train_aug

        state = torch.load(ckpt_path, map_location=device)
        expected_n_num = int(state["num_embed.w1"].shape[1])

        model = RealMLP(
            input_dim=int(expected_n_num * 5 + sum([c if c <= 8 els
            output_dim=2,
            cat_dims=cat_dims_global,
            n_numerical=expected_n_num,
            n_ens=n_ens,
            embed_dim=embed_dim,
            device=device,
        ).to(device)
        model.load_state_dict(state)
        model.eval()

        cache.append({
            "fold": fold,
            "train_idx": train_idx,
            "valid_idx": valid_idx,
            "te": te,
            "rssc": rssc,
            "model": model,
            "device": device,
        })
    return cache


def predict_realmlp_custom(raw_df, realmlp_bundle, cfg, fold_cache,
```

```
    _, X_num_custom, X_cat_custom = prepare_custom_realmlp_df(raw_d
    cols_to_encode = realmlp_bundle["cols_to_encode"]

    preds = []
    member_preds = []
    for item in fold_cache:
        te = item["te"]
        rssc = item["rssc"]
        model = item["model"]
        device = item["device"]

        if len(cols_to_encode) > 0:
            X_cat_encoded = te.transform(X_cat_custom[:, cols_to_en
        else:
            X_cat_encoded = np.zeros((len(X_num_custom), 0), dtype=

        X_num_aug = np.concatenate([X_num_custom, X_cat_encoded], a
        X_num_aug = rssc.transform(X_num_aug)

        X_num_t = torch.as_tensor(X_num_aug, dtype=torch.float32, d
        X_cat_t = torch.as_tensor(X_cat_custom, dtype=torch.long, d

        fold_probs = []
        with torch.no_grad():
            for start in range(0, len(raw_df), eval_bs):
                probs = model(X_num_t[start:start + eval_bs], X_cat
                fold_probs.append(probs.detach().cpu().numpy())
        fold_probs = np.concatenate(fold_probs, axis=0)
        preds.append(fold_probs)

        # Member-level predictions from the ensemble dimension
        with torch.no_grad():
            member_block = []
            for start in range(0, len(raw_df), eval_bs):
                prob_full = model(X_num_t[start:start + eval_bs], X
                member_block.append(prob_full.detach().cpu().numpy(
        member_block = np.concatenate(member_block, axis=0)
        member_preds.append(member_block)

    preds = np.vstack(preds)
    mean_pred = preds.mean(axis=0)

    member_preds = np.stack(member_preds, axis=0)   # fold x row x
    flat_member_preds = member_preds.transpose(1, 0, 2).reshape(len

    return {
        "mean_pred": mean_pred,
        "fold_pred_matrix": preds,
        "member_pred_matrix": flat_member_preds,
        "custom_X_num": X_num_custom,
        "custom_X_cat": X_cat_custom,
    }


fold_cache = build_realmlp_fold_cache(realmlp_bundle, cfg)
print("Cached folds:", len(fold_cache))
```

```
Cached folds: 10
```

In [14]:
```
def normalise_realmlp_fold_cache(fold_cache):
    normalised = []

    for i, fold_obj in enumerate(fold_cache):
        keys = set(fold_obj.keys())

        model = None
        target_encoder = None
        rssc = None

        for k in ["model", "net", "estimator"]:
            if k in fold_obj:
                model = fold_obj[k]
                break

        for k in ["target_encoder", "te", "target_enc", "encoder"]:
            if k in fold_obj:
                target_encoder = fold_obj[k]
                break

        for k in ["rssc", "scaler", "transform", "num_transform"]:
            if k in fold_obj:
                rssc = fold_obj[k]
                break

        if model is None:
            raise KeyError(
                f"Fold cache entry {i} is missing a model-like key.
            )
        if target_encoder is None:
            raise KeyError(
                f"Fold cache entry {i} is missing a target-encoder-
            )
        if rssc is None:
            raise KeyError(
                f"Fold cache entry {i} is missing an rssc/scaler-li
            )

        normalised.append({
            "model": model,
            "target_encoder": target_encoder,
            "rssc": rssc,
        })

    return normalised


fold_cache = normalise_realmlp_fold_cache(fold_cache)
print(fold_cache[0].keys())
```

```
dict_keys(['model', 'target_encoder', 'rssc'])
```

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In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

5. Permutation importance on the replayed competition train set

For the neural model, permutation importance is a robust first-line global explanation because it does not require a tree-specific decomposition and respects the actual fitted predictor.

In [33]:
```
from sklearn.metrics import roc_auc_score

def _fold_get(fold_obj, *names):
    for name in names:
        if name in fold_obj:
            return fold_obj[name]
    raise KeyError(f"None of {names} found in fold cache keys: {sor


def _build_realmlp_cached_fold_inputs(
    realmlp_bundle,
    fold_cache,
    row_idx=None,
):
    X_num = realmlp_bundle["X_num"].astype(np.float32)
    X_cat = realmlp_bundle["X_cat"].astype(np.int64)
    y = realmlp_bundle["y_nn"]
    cols_to_encode = realmlp_bundle["cols_to_encode"]

    if row_idx is None:
        row_idx = np.arange(X_num.shape[0])

    X_num = X_num[row_idx].copy()
    X_cat = X_cat[row_idx].copy()
    y = y[row_idx].copy()

    cached = []
    for fold_obj in fold_cache:
        te = _fold_get(fold_obj, "target_encoder", "te", "target_en
        rssc = _fold_get(fold_obj, "rssc", "scaler", "transform", "
        model = _fold_get(fold_obj, "model", "net", "estimator")

        if len(cols_to_encode) > 0:
            X_cat_encoded = te.transform(X_cat[:, cols_to_encode]).
        else:
            X_cat_encoded = np.zeros((X_num.shape[0], 0), dtype=np.

        X_num_aug = np.concatenate([X_num, X_cat_encoded], axis=1).
        X_num_aug = rssc.transform(X_num_aug).astype(np.float32)

        cached.append({
```

```
            "model": model,
            "X_num_aug": X_num_aug,
            "X_cat": X_cat,
        })

    return cached, y, row_idx


def _predict_realmlp_from_cached(
    cached_fold_inputs,
    eval_bs=32768,
):
    all_fold_preds = []

    for fold_obj in cached_fold_inputs:
        model = fold_obj["model"]
        X_num_aug = fold_obj["X_num_aug"]
        X_cat = fold_obj["X_cat"]

        device = next(model.parameters()).device

        X_num_t = torch.as_tensor(X_num_aug, dtype=torch.float32, d
        X_cat_t = torch.as_tensor(X_cat, dtype=torch.long, device=d

        preds = []
        model.eval()
        with torch.no_grad():
            for start in range(0, X_num_t.shape[0], eval_bs):
                batch_num = X_num_t[start:start + eval_bs]
                batch_cat = X_cat_t[start:start + eval_bs]
                probs = model(batch_num, batch_cat).mean(dim=1)[:,
                preds.append(probs.detach().cpu().numpy())

        all_fold_preds.append(np.concatenate(preds, axis=0))

    all_fold_preds = np.column_stack(all_fold_preds)
    mean_pred = all_fold_preds.mean(axis=1)
    std_pred = all_fold_preds.std(axis=1)

    return {
        "member_preds": all_fold_preds,
        "mean_pred": mean_pred,
        "std_pred": std_pred,
    }


def realmlp_numeric_permutation_importance_fast(
    realmlp_bundle,
    fold_cache,
    n_repeats=2,
    sample_n=120_000,
    random_state=67,
    eval_bs=32768,
):
    X_num_full = realmlp_bundle["X_num"]
    feature_names = list(realmlp_bundle["NUMS"])
```

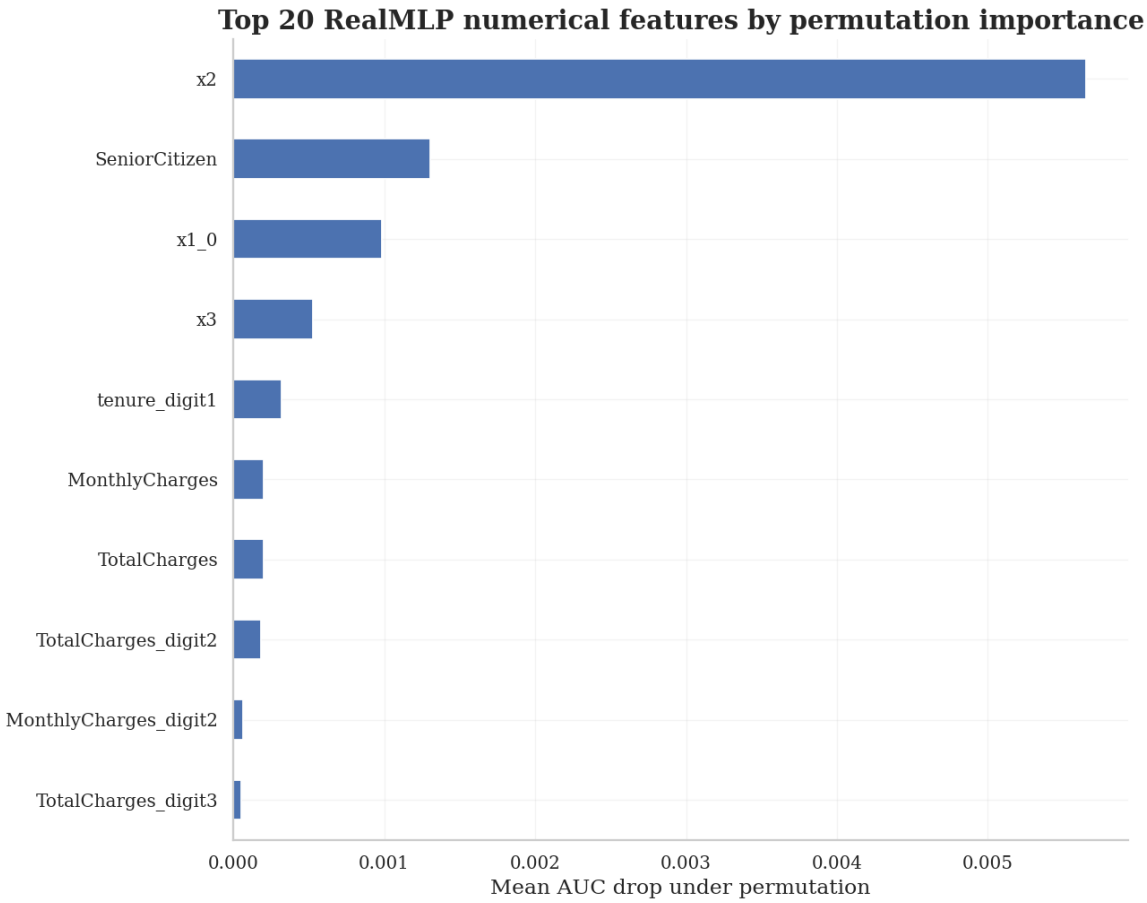
```
    rng = np.random.RandomState(random_state)

    if sample_n is not None and sample_n < X_num_full.shape[0]:
        row_idx = rng.choice(X_num_full.shape[0], size=sample_n, re
        row_idx = np.sort(row_idx)
    else:
        row_idx = np.arange(X_num_full.shape[0])

    cached_fold_inputs, y, row_idx = _build_realmlp_cached_fold_inp
        realmlp_bundle=realmlp_bundle,
        fold_cache=fold_cache,
        row_idx=row_idx,
    )

    baseline = _predict_realmlp_from_cached(
        cached_fold_inputs,
        eval_bs=eval_bs,
    )["mean_pred"]
    baseline_auc = roc_auc_score(y, baseline)

    # number of original numerical features before target-encoded c
    n_model_num_features = realmlp_bundle["X_num"].shape[1]

    rows = []
    for j, feat in enumerate(feature_names):
        aucs = []

        for _ in range(n_repeats):
            perm_idx = rng.permutation(len(y))

            permuted_cached = []
            for fold_obj in cached_fold_inputs:
                X_num_aug_perm = fold_obj["X_num_aug"].copy()
                # permute only the original numerical column j; kee
                X_num_aug_perm[:, j] = X_num_aug_perm[perm_idx, j]

                permuted_cached.append({
                    "model": fold_obj["model"],
                    "X_num_aug": X_num_aug_perm,
                    "X_cat": fold_obj["X_cat"],
                })

            pred_perm = _predict_realmlp_from_cached(
                permuted_cached,
                eval_bs=eval_bs,
            )["mean_pred"]

            aucs.append(roc_auc_score(y, pred_perm))

        aucs = np.asarray(aucs, dtype=float)
        rows.append({
            "feature": feat,
            "baseline_auc": baseline_auc,
            "perm_auc_mean": aucs.mean(),
            "perm_auc_std": aucs.std(ddof=0),
            "mean_importance": baseline_auc - aucs.mean(),
```

```
            "std_importance": aucs.std(ddof=0),
        })

        print(
            f"{j+1:>3d}/{len(feature_names)}  {feat:<30s}  "
            f"AUC drop = {baseline_auc - aucs.mean():.6f}"
        )

    out = (
        pd.DataFrame(rows)
        .sort_values("mean_importance", ascending=False)
        .reset_index(drop=True)
    )
    return out, baseline_auc, row_idx


perm_imp, baseline_auc, row_idx = realmlp_numeric_permutation_impor
    realmlp_bundle=realmlp_bundle,
    fold_cache=fold_cache,
    n_repeats=2,          # was 5
    sample_n=120_000,     # set None for full data
    random_state=cfg.seed,
    eval_bs=32768,        # larger batches = faster if memory allow
)

print(f"Baseline sampled OOF AUC from cached replay path: {baseline
print(f"Rows used: {len(row_idx):,}")
display(perm_imp.head(25))

ax = perm_imp.head(10).sort_values("mean_importance").plot(
    x="feature",
    y="mean_importance",
    kind="barh",
    legend=False,
    figsize=(10, 8),
)
ax.set_title("Top 20 RealMLP numerical features by permutation impo
ax.set_xlabel("Mean AUC drop under permutation")
ax.set_ylabel("")
savefig(cfg, "realmlp_02_permutation_importance.png")
plt.show()
```

```
  1/22  SeniorCitizen                   AUC drop = 0.001306
  2/22  tenure                          AUC drop = 0.000053
  3/22  MonthlyCharges                  AUC drop = 0.000202
  4/22  TotalCharges                    AUC drop = 0.000198
  5/22  tenure_digit0                   AUC drop = 0.000022
  6/22  tenure_digit1                   AUC drop = 0.000319
  7/22  MonthlyCharges_digit-2          AUC drop = 0.000012
  8/22  MonthlyCharges_digit-1          AUC drop = 0.000013
  9/22  MonthlyCharges_digit0           AUC drop = 0.000009
 10/22  MonthlyCharges_digit1           AUC drop = 0.000029
 11/22  MonthlyCharges_digit2           AUC drop = 0.000066
 12/22  TotalCharges_digit-2            AUC drop = 0.000010
 13/22  TotalCharges_digit-1            AUC drop = 0.000013
 14/22  TotalCharges_digit0             AUC drop = 0.000014
 15/22  TotalCharges_digit1             AUC drop = 0.000045
 16/22  TotalCharges_digit2             AUC drop = 0.000185
 17/22  TotalCharges_digit3             AUC drop = 0.000055
 18/22  x1_0                            AUC drop = 0.000986
 19/22  x1_1                            AUC drop = 0.000051
 20/22  x2                              AUC drop = 0.005648
 21/22  x3                              AUC drop = 0.000530
 22/22  x4                              AUC drop = 0.000030
Baseline sampled OOF AUC from cached replay path: 0.918749
Rows used: 120,000
```

	feature	baseline_auc	perm_auc_mean	perm_auc_std	mea
0	x2	0.918749	0.913101	1.381999e-04	
1	SeniorCitizen	0.918749	0.917442	3.250889e-05	
2	x1_0	0.918749	0.917763	9.978155e-06	
3	x3	0.918749	0.918219	9.073667e-06	
4	tenure_digit1	0.918749	0.918430	4.806324e-05	
5	MonthlyCharges	0.918749	0.918547	1.387151e-05	
6	TotalCharges	0.918749	0.918551	1.026089e-05	
7	TotalCharges_digit2	0.918749	0.918564	1.885185e-05	
8	MonthlyCharges_digit2	0.918749	0.918683	1.561111e-05	
9	TotalCharges_digit3	0.918749	0.918694	9.975575e-06	
10	tenure	0.918749	0.918696	8.026072e-06	
11	x1_1	0.918749	0.918698	5.403114e-06	
12	TotalCharges_digit1	0.918749	0.918704	1.795282e-07	
13	x4	0.918749	0.918719	6.388782e-06	
14	MonthlyCharges_digit1	0.918749	0.918719	4.799625e-06	
15	tenure_digit0	0.918749	0.918727	1.408537e-06	
16	TotalCharges_digit0	0.918749	0.918735	1.842819e-06	
17	MonthlyCharges_digit-1	0.918749	0.918735	3.261082e-07	
18	TotalCharges_digit-1	0.918749	0.918736	2.908277e-06	
19	MonthlyCharges_digit-2	0.918749	0.918737	1.989796e-06	
20	TotalCharges_digit-2	0.918749	0.918739	2.543564e-07	
21	MonthlyCharges_digit0	0.918749	0.918739	1.537353e-06	

```
saved: analysis_outputs\realmlp_02_permutation_importance.png
```



In [20]:
```
raw_train_no_id = realmlp_bundle["train_nn"].drop(columns=[cfg.id_c
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

6. Local finite-difference sensitivity

We approximate local sensitivity of the probability surface with finite differences:

$$S_j(x) \approx \frac{p(x + he_j) - p(x - he_j)}{2h}.$$

This is not a gradient through the full preprocessing graph. It is a practical response derivative on the reconstructed end-to-end predictor.

In [24]:
```
def finite_difference_sensitivity_arrays(
    realmlp_bundle,
    fold_cache,
    features,
    n_rows=1500,
    row_idx=None,
```

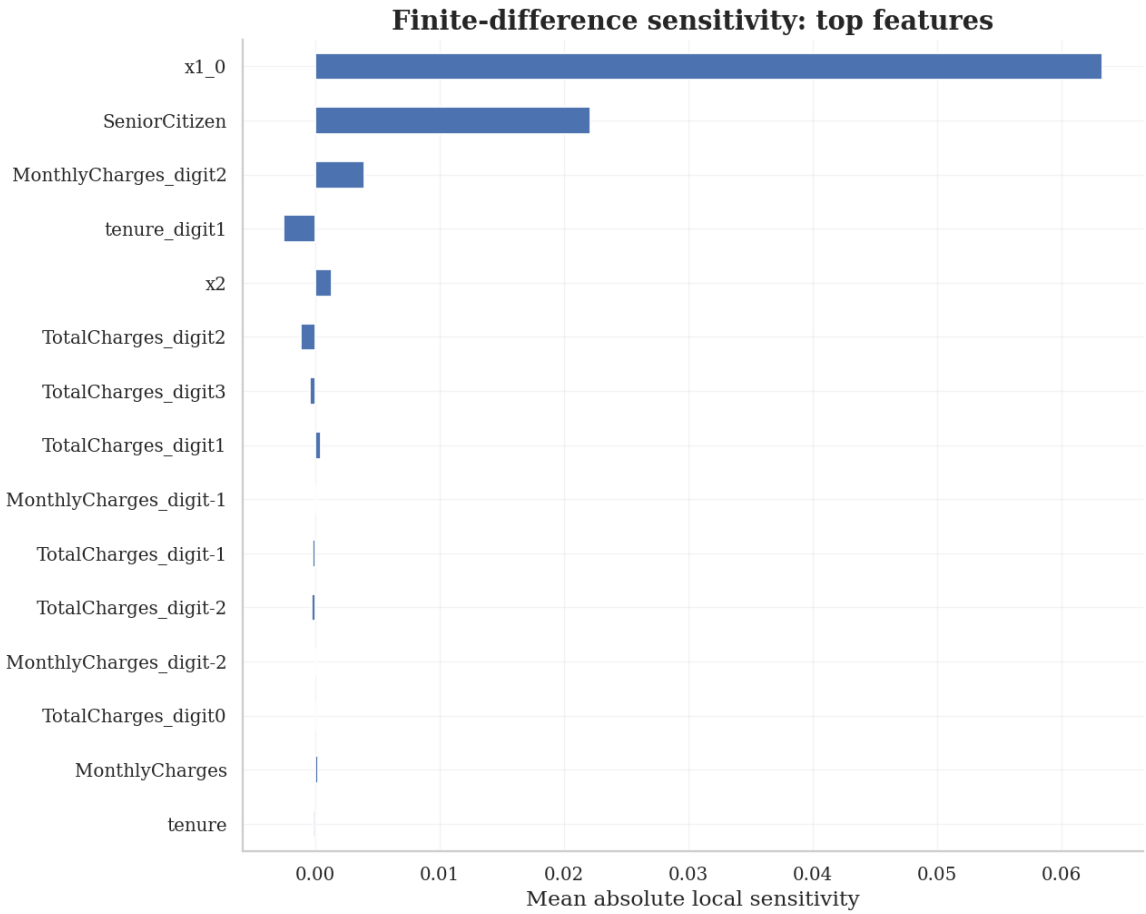
```
    eval_bs=32768,
):
    X_num_full = realmlp_bundle["X_num"].astype(np.float32)
    X_cat_full = realmlp_bundle["X_cat"].astype(np.int64)
    feature_names = list(realmlp_bundle["NUMS"])

    feat_to_idx = {f: i for i, f in enumerate(feature_names)}
    valid_features = [f for f in features if f in feat_to_idx]

    if row_idx is None:
        row_idx = np.arange(min(n_rows, X_num_full.shape[0]))
    else:
        row_idx = np.asarray(row_idx)[:n_rows]

    X_num = X_num_full[row_idx].copy()
    X_cat = X_cat_full[row_idx].copy()

    base_out = predict_realmlp_from_arrays(
        X_num, X_cat, realmlp_bundle, fold_cache, eval_bs=eval_bs
    )
    base_pred = base_out["mean_pred"]

    rows = []
    for feat in valid_features:
        j = feat_to_idx[feat]
        col = X_num[:, j].astype(np.float64)

        sd = float(np.std(col, ddof=0))
        step = max(sd * 0.05, 1e-3)

        X_num_plus = X_num.copy()
        X_num_minus = X_num.copy()
        X_num_plus[:, j] = X_num_plus[:, j] + step
        X_num_minus[:, j] = X_num_minus[:, j] - step

        plus_pred = predict_realmlp_from_arrays(
            X_num_plus, X_cat, realmlp_bundle, fold_cache, eval_bs=
        )["mean_pred"]
        minus_pred = predict_realmlp_from_arrays(
            X_num_minus, X_cat, realmlp_bundle, fold_cache, eval_bs
        )["mean_pred"]

        sens = (plus_pred - minus_pred) / (2.0 * step)

        rows.append({
            "feature": feat,
            "mean_abs_sensitivity": float(np.mean(np.abs(sens))),
            "mean_signed_sensitivity": float(np.mean(sens)),
            "step": float(step),
            "base_pred_mean": float(np.mean(base_pred)),
        })

        print(
            f"{len(rows):>2d}/{len(valid_features)}  {feat:<30s}  "
            f"|mean sens| = {np.mean(np.abs(sens)):.6f}"
        )
```

```
    return pd.DataFrame(rows).sort_values("mean_abs_sensitivity", a


numeric_candidates = list(realmlp_bundle["NUMS"])[:20]

sens_df = finite_difference_sensitivity_arrays(
    realmlp_bundle=realmlp_bundle,
    fold_cache=fold_cache,
    features=numeric_candidates,
    n_rows=1500,
    eval_bs=32768,
)

display(sens_df.head(20))

ax = sens_df.head(15).sort_values("mean_abs_sensitivity").plot(
    x="feature",
    y="mean_signed_sensitivity",
    kind="barh",
    legend=False,
    figsize=(10, 8),
)
ax.set_title("Finite-difference sensitivity: top features")
ax.set_xlabel("Mean absolute local sensitivity")
ax.set_ylabel("")
savefig(cfg, "realmlp_03_finite_difference_sensitivity.png")
plt.show()
```

```
 1/20  SeniorCitizen                    |mean sens| = 0.022135
 2/20  tenure                           |mean sens| = 0.000223
 3/20  MonthlyCharges                   |mean sens| = 0.000229
 4/20  TotalCharges                     |mean sens| = 0.000003
 5/20  tenure_digit0                    |mean sens| = 0.000140
 6/20  tenure_digit1                    |mean sens| = 0.002566
 7/20  MonthlyCharges_digit-2           |mean sens| = 0.000292
 8/20  MonthlyCharges_digit-1           |mean sens| = 0.000427
 9/20  MonthlyCharges_digit0            |mean sens| = 0.000200
10/20  MonthlyCharges_digit1            |mean sens| = 0.000220
11/20  MonthlyCharges_digit2            |mean sens| = 0.003959
12/20  TotalCharges_digit-2             |mean sens| = 0.000306
13/20  TotalCharges_digit-1             |mean sens| = 0.000342
14/20  TotalCharges_digit0              |mean sens| = 0.000290
15/20  TotalCharges_digit1              |mean sens| = 0.000673
16/20  TotalCharges_digit2              |mean sens| = 0.001161
17/20  TotalCharges_digit3              |mean sens| = 0.000694
18/20  x1_0                             |mean sens| = 0.109892
19/20  x1_1                             |mean sens| = 0.000152
20/20  x2                               |mean sens| = 0.001274
```

	feature	mean_abs_sensitivity	mean_signed_sensitivity	
0	x1_0	0.109892	0.063291	0.
1	SeniorCitizen	0.022135	0.022135	0.
2	MonthlyCharges_digit2	0.003959	0.003959	0.
3	tenure_digit1	0.002566	-0.002566	0.
4	x2	0.001274	0.001268	1.
5	TotalCharges_digit2	0.001161	-0.001143	0.
6	TotalCharges_digit3	0.000694	-0.000433	0
7	TotalCharges_digit1	0.000673	0.000463	0.
8	MonthlyCharges_digit-1	0.000427	0.000064	0.
9	TotalCharges_digit-1	0.000342	-0.000215	0
10	TotalCharges_digit-2	0.000306	-0.000301	0.
11	MonthlyCharges_digit-2	0.000292	0.000061	0.
12	TotalCharges_digit0	0.000290	0.000002	0
13	MonthlyCharges	0.000229	0.000221	1.
14	tenure	0.000223	-0.000173	1.
15	MonthlyCharges_digit1	0.000220	-0.000177	0.
16	MonthlyCharges_digit0	0.000200	0.000063	0.
17	x1_1	0.000152	-0.000108	1.
18	tenure_digit0	0.000140	-0.000051	0
19	TotalCharges	0.000003	-0.000001	116.

```
saved: analysis_outputs\realmlp_03_finite_difference_sensitivity.png
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

7. Grouped categorical ablation

For the categorical branch we use grouped ablations. A group is neutralised by setting its columns to a reference category profile and measuring the induced probability shift.

This directly answers which categorical families the neural model is leaning on most.

In [25]:
```
def categorical_group_ablation_arrays(
    realmlp_bundle,
    fold_cache,
    cat_groups,
    n_rows=3000,
    eval_bs=32768,
):
    X_num_full = realmlp_bundle["X_num"].astype(np.float32)
    X_cat_full = realmlp_bundle["X_cat"].astype(np.int64)
    y = realmlp_bundle["y_nn"]
```

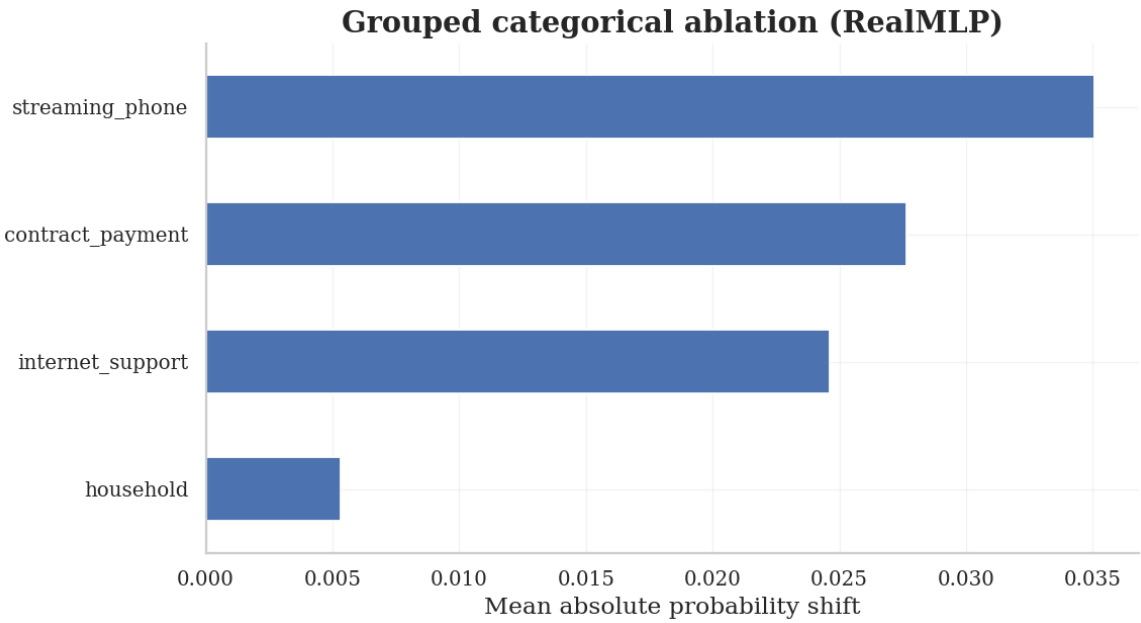
```
    CATEGORY = realmlp_bundle["CATEGORY"]

    cat_to_idx = {c: i for i, c in enumerate(CATEGORY)}

    row_idx = np.arange(min(n_rows, X_num_full.shape[0]))
    X_num = X_num_full[row_idx].copy()
    X_cat = X_cat_full[row_idx].copy()

    base_pred = predict_realmlp_from_arrays(
        X_num, X_cat, realmlp_bundle, fold_cache, eval_bs=eval_bs
    )["mean_pred"]

    ablation_rows = []

    for group_name, cols in cat_groups.items():
        idxs = [cat_to_idx[c] for c in cols if c in cat_to_idx]
        if not idxs:
            continue

        X_cat_abl = X_cat.copy()

        # replace with most frequent category (mode in encoded spac
        for j in idxs:
            values, counts = np.unique(X_cat[:, j], return_counts=T
            mode_val = values[np.argmax(counts)]
            X_cat_abl[:, j] = mode_val

        abl_pred = predict_realmlp_from_arrays(
            X_num, X_cat_abl, realmlp_bundle, fold_cache, eval_bs=e
        )["mean_pred"]

        delta = base_pred - abl_pred

        ablation_rows.append({
            "group": group_name,
            "mean_abs_delta": float(np.mean(np.abs(delta))),
            "mean_signed_delta": float(np.mean(delta)),
            "max_abs_delta": float(np.max(np.abs(delta))),
        })

        print(
            f"{group_name:<20s}  |Δ| = {np.mean(np.abs(delta)):.6f}
        )

    return pd.DataFrame(ablation_rows).sort_values("mean_abs_delta"


cat_groups = {
    "contract_payment": ["Contract", "PaymentMethod", "PaperlessBil
    "internet_support": ["InternetService", "OnlineSecurity", "Onli
    "streaming_phone": ["PhoneService", "MultipleLines", "Streaming
    "household": ["gender", "SeniorCitizen", "Partner", "Dependents
}

ablation_df = categorical_group_ablation_arrays(
```

```
    realmlp_bundle=realmlp_bundle,
    fold_cache=fold_cache,
    cat_groups=cat_groups,
    n_rows=3000,
    eval_bs=32768,
)

display(ablation_df)

ax = ablation_df.sort_values("mean_abs_delta").plot(
    x="group",
    y="mean_abs_delta",
    kind="barh",
    legend=False,
    figsize=(9, 5),
)
ax.set_title("Grouped categorical ablation (RealMLP)")
ax.set_xlabel("Mean absolute probability shift")
ax.set_ylabel("")
savefig(cfg, "realmlp_04_grouped_ablation.png")
plt.show()
```

```
contract_payment      |Δ| = 0.027666
internet_support      |Δ| = 0.024606
streaming_phone       |Δ| = 0.035071
household             |Δ| = 0.005301
```

	group	mean_abs_delta	mean_signed_delta	max_abs_delta
2	streaming_phone	0.035071	0.034967	0.223998
0	contract_payment	0.027666	-0.027666	0.198302
1	internet_support	0.024606	-0.024606	0.317824
3	household	0.005301	-0.004702	0.110057

```
saved: analysis_outputs\realmlp_04_grouped_ablation.png
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

8. Conditional decision surfaces

For selected numerical feature pairs, we visualise two-dimensional conditional slices:

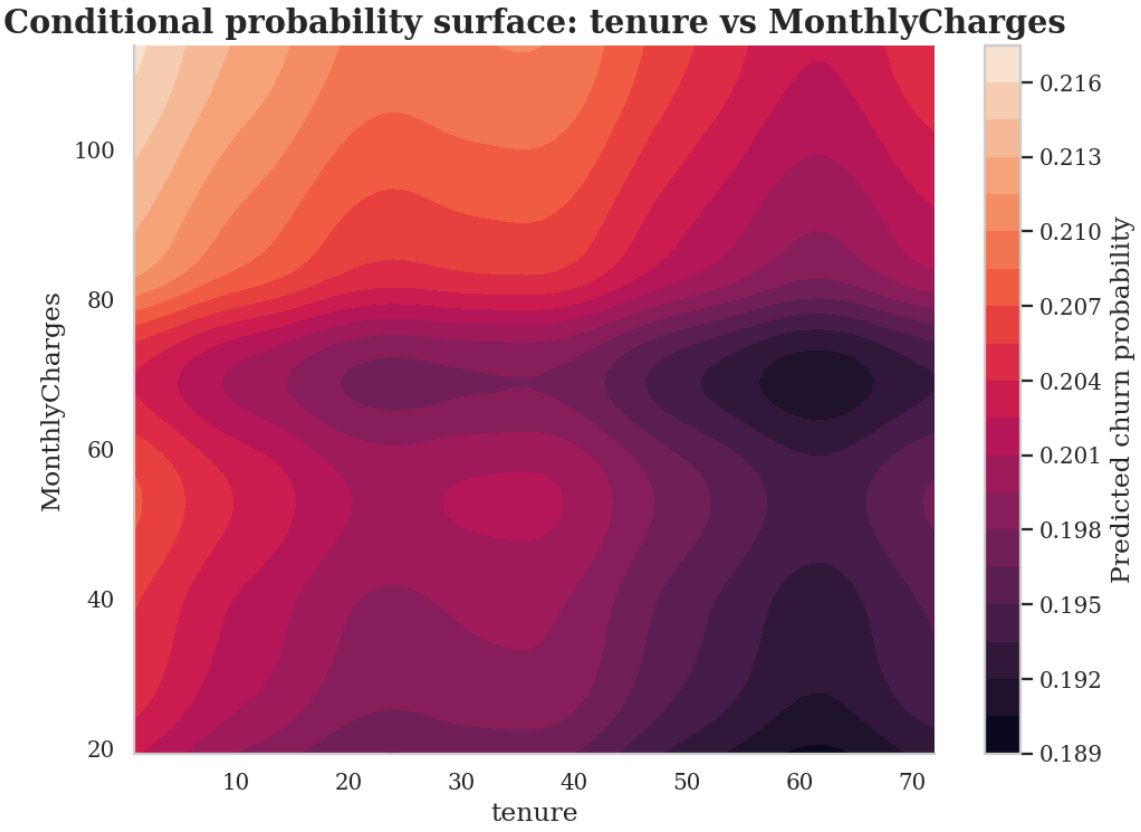
$$(x_i, x_j) \mapsto p_{\mathrm{mlp}}(x_i, x_j, \tilde{x}_{-ij}).$$

Compared with the tree model, these surfaces should be smoother and more curved if the neural representation is interpolating rather than thresholding.

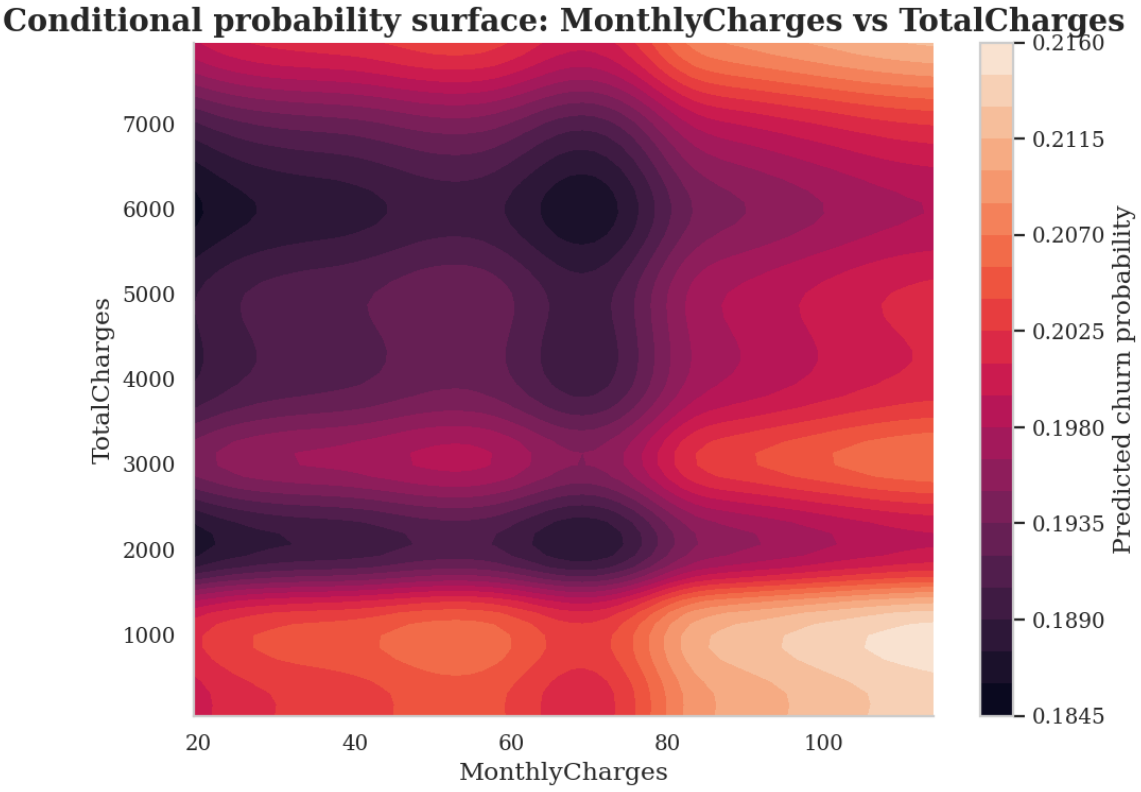
In [28]:
```
def plot_realmlp_surface_arrays(
    realmlp_bundle,
    fold_cache,
    feature_x,
    feature_y,
    grid_n=60,
    q_low=0.02,
    q_high=0.98,
    eval_bs=32768,
):
    feature_names = list(realmlp_bundle["NUMS"])
    feat_to_idx = {f: i for i, f in enumerate(feature_names)}

    if feature_x not in feat_to_idx or feature_y not in feat_to_idx
        raise KeyError(
            f"Both features must be in realmlp_bundle['NUMS']. "
            f"Missing: {[f for f in [feature_x, feature_y] if f not
        )

    X_num_full = realmlp_bundle["X_num"].astype(np.float32)
    X_cat_full = realmlp_bundle["X_cat"].astype(np.int64)

    jx = feat_to_idx[feature_x]
    jy = feat_to_idx[feature_y]

    x_col = X_num_full[:, jx].astype(np.float64)
    y_col = X_num_full[:, jy].astype(np.float64)

    xs = np.linspace(np.quantile(x_col, q_low), np.quantile(x_col,
    ys = np.linspace(np.quantile(y_col, q_low), np.quantile(y_col,

    # reference point = column medians in numeric input space,
    # modal category codes in categorical input space
    ref_num = np.median(X_num_full, axis=0).astype(np.float32)
    ref_cat = np.zeros(X_cat_full.shape[1], dtype=np.int64)
    for j in range(X_cat_full.shape[1]):
        vals, counts = np.unique(X_cat_full[:, j], return_counts=Tr
        ref_cat[j] = vals[np.argmax(counts)]
```

```
    # build grid
    X_num_grid = np.repeat(ref_num[None, :], grid_n * grid_n, axis=
    X_cat_grid = np.repeat(ref_cat[None, :], grid_n * grid_n, axis=

    # write the 2D slice
    xx, yy = np.meshgrid(xs, ys)
    X_num_grid[:, jx] = xx.ravel().astype(np.float32)
    X_num_grid[:, jy] = yy.ravel().astype(np.float32)

    out = predict_realmlp_from_arrays(
        X_num_grid,
        X_cat_grid,
        realmlp_bundle,
        fold_cache,
        eval_bs=eval_bs,
    )
    z = out["mean_pred"].reshape(len(ys), len(xs))

    fig, ax = plt.subplots(figsize=(8, 6))
    cf = ax.contourf(xs, ys, z, levels=20)
    ax.contour(xs, ys, z, levels=[0.5], linewidths=2)
    ax.set_xlabel(feature_x)
    ax.set_ylabel(feature_y)
    ax.set_title(f"Conditional probability surface: {feature_x} vs
    plt.colorbar(cf, ax=ax, label="Predicted churn probability")
    return fig, ax


surface_features = [
    ("tenure", "MonthlyCharges"),
    ("MonthlyCharges", "TotalCharges"),
    ("tenure", "x2"),
    ("x1_0", "x2"),
]

for a, b in surface_features:
    if a in realmlp_bundle["NUMS"] and b in realmlp_bundle["NUMS"]:
        fig, ax = plot_realmlp_surface_arrays(
            realmlp_bundle=realmlp_bundle,
            fold_cache=fold_cache,
            feature_x=a,
            feature_y=b,
            grid_n=60,
            q_low=0.02,
            q_high=0.98,
            eval_bs=32768,
        )
        savefig(cfg, f"realmlp_surface_{a}_{b}.png".replace("/", "_
        plt.show()
```

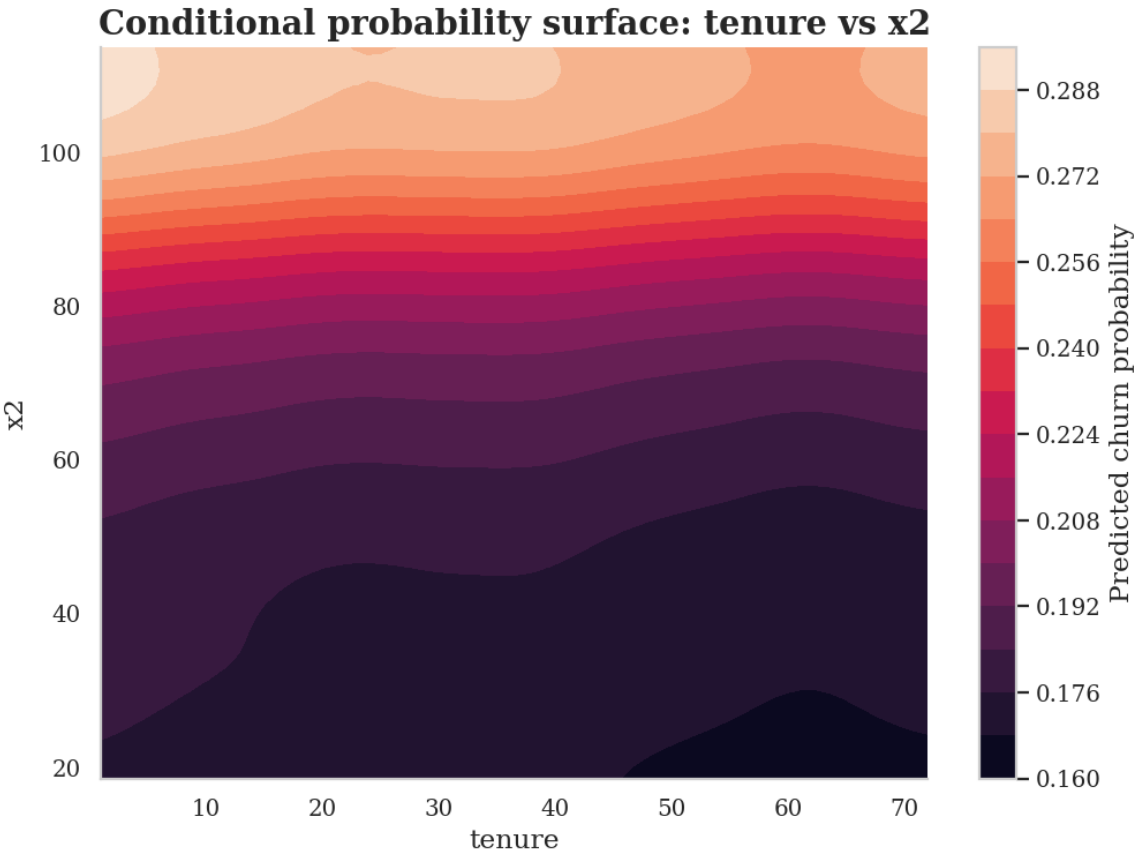
```
saved: analysis_outputs\realmlp_surface_tenure_MonthlyCharges.png
```



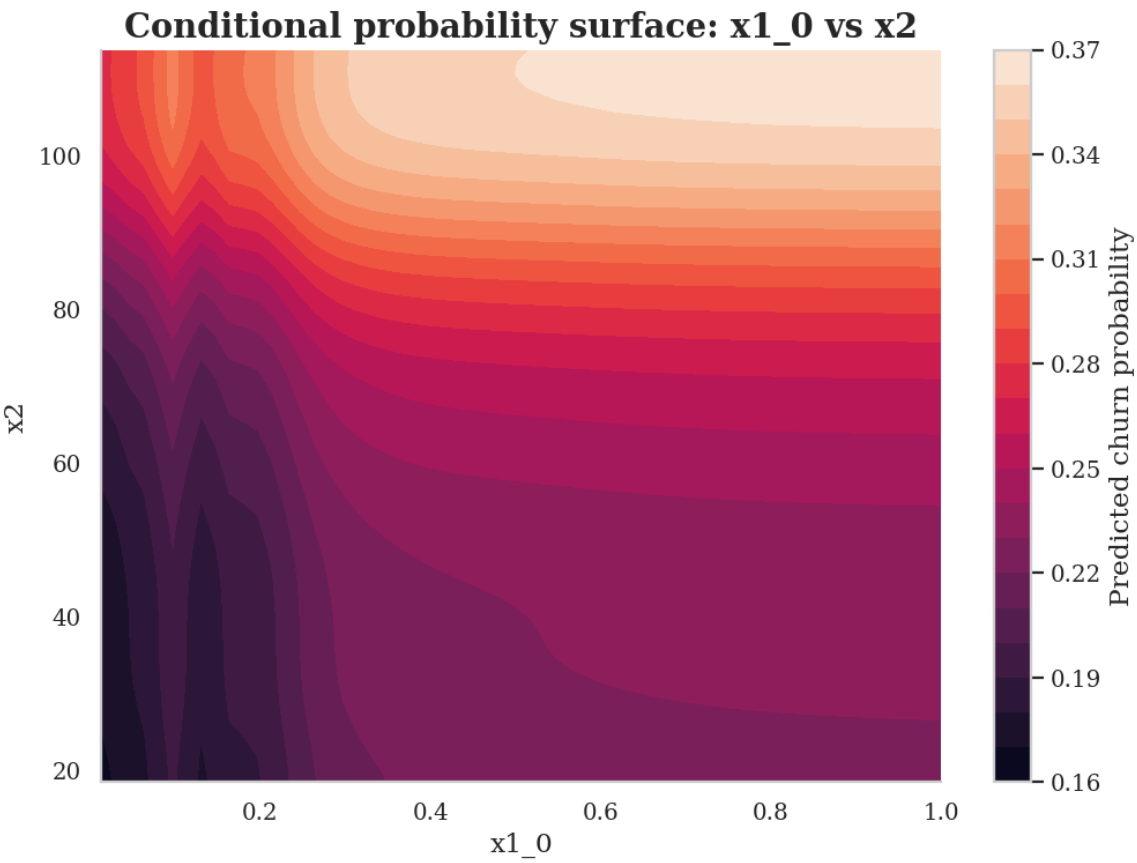

```
saved: analysis_outputs\realmlp_surface_MonthlyCharges_TotalCharges.
png
```




```
saved: analysis_outputs\realmlp_surface_tenure_x2.png
```




```
saved: analysis_outputs\realmlp_surface_x1_0_x2.png
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this

function such as gradients, perturbations, or expectations.

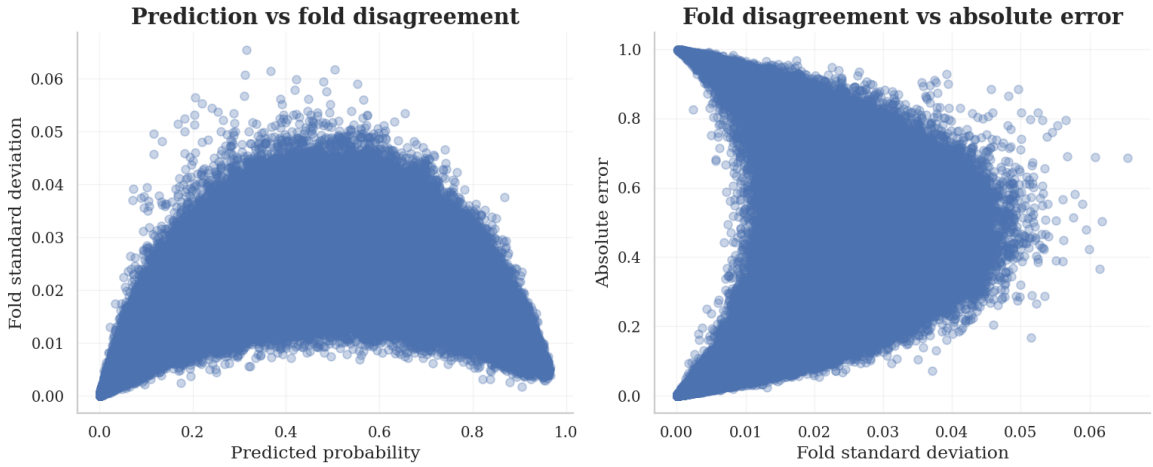
10. Disagreement and uncertainty structure

Fold-model averaging gives us a natural uncertainty proxy: disagreement across fold-members.

We inspect where disagreement is highest and whether high disagreement aligns with higher error.

In [30]:
```
# full train-side member prediction matrix across all fold models
train_member_out = predict_realmlp_from_arrays(
    realmlp_bundle["X_num"].astype(np.float32),
    realmlp_bundle["X_cat"].astype(np.int64),
    realmlp_bundle,
    fold_cache,
    eval_bs=32768,
)

oof_fold_pred = train_member_out["member_preds"]

analysis_train["fold_std"] = oof_fold_pred.std(axis=1)
analysis_train["fold_range"] = oof_fold_pred.max(axis=1) - oof_fold

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

axes[0].scatter(analysis_train["realmlp_pred"], analysis_train["fol
axes[0].set_title("Prediction vs fold disagreement")
axes[0].set_xlabel("Predicted probability")
axes[0].set_ylabel("Fold standard deviation")

axes[1].scatter(analysis_train["fold_std"], analysis_train["abs_err
axes[1].set_title("Fold disagreement vs absolute error")
axes[1].set_xlabel("Fold standard deviation")
axes[1].set_ylabel("Absolute error")

plt.tight_layout()
savefig(cfg, "realmlp_05_disagreement_vs_error.png")
plt.show()

uncertainty_summary = analysis_train.groupby(
    pd.qcut(analysis_train["fold_std"], q=5, duplicates="drop")
).agg(
    mean_pred=("realmlp_pred", "mean"),
    observed_rate=("y_true", "mean"),
    mean_abs_error=("abs_error", "mean"),
    n=("y_true", "size"),
)
display(uncertainty_summary)
```

```
saved: analysis_outputs\realmlp_05_disagreement_vs_error.png
```



fold_std	mean_pred	observed_rate	mean_abs_error	n
(-0.0009382, 0.000666]	0.003392	0.001750	0.005128	118839
(0.000666, 0.00317]	0.012918	0.011335	0.023767	118839
(0.00317, 0.0136]	0.147528	0.151021	0.138210	118838
(0.0136, 0.0233]	0.411039	0.442700	0.354451	118839
(0.0233, 0.0656]	0.474269	0.519232	0.449057	118839

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

11. Local case studies

We create representative local diagnostics for:

- extreme TP,
- extreme TN,
- high-confidence FP,
- low-score FN.

For the neural model, the most honest local explanation is a local perturbation report:

- base prediction,
- top finite-difference sensitivities for numerical variables,
- effect of grouped categorical ablations for that row.

In [32]:
```
def local_numeric_sensitivity_for_row_arrays(
```

```
    row_index,
    realmlp_bundle,
    fold_cache,
    features,
    n_top=12,
    eval_bs=32768,
):
    X_num_full = realmlp_bundle["X_num"].astype(np.float32)
    X_cat_full = realmlp_bundle["X_cat"].astype(np.int64)
    feature_names = list(realmlp_bundle["NUMS"])
    feat_to_idx = {f: i for i, f in enumerate(feature_names)}

    valid_features = [f for f in features if f in feat_to_idx]

    X_num_row = X_num_full[[row_index]].copy()
    X_cat_row = X_cat_full[[row_index]].copy()

    base = predict_realmlp_from_arrays(
        X_num_row,
        X_cat_row,
        realmlp_bundle,
        fold_cache,
        eval_bs=eval_bs,
    )["mean_pred"][0]

    rows = []
    for feat in valid_features:
        j = feat_to_idx[feat]

        col_all = X_num_full[:, j].astype(np.float64)
        step = max(float(np.std(col_all, ddof=0)) * 0.05, 1e-3)

        X_num_plus = X_num_row.copy()
        X_num_minus = X_num_row.copy()
        X_num_plus[0, j] = X_num_plus[0, j] + step
        X_num_minus[0, j] = X_num_minus[0, j] - step

        p_plus = predict_realmlp_from_arrays(
            X_num_plus,
            X_cat_row,
            realmlp_bundle,
            fold_cache,
            eval_bs=eval_bs,
        )["mean_pred"][0]

        p_minus = predict_realmlp_from_arrays(
            X_num_minus,
            X_cat_row,
            realmlp_bundle,
            fold_cache,
            eval_bs=eval_bs,
        )["mean_pred"][0]

        sens = (p_plus - p_minus) / (2.0 * step)

        rows.append({
```

```
            "feature": feat,
            "local_abs_sensitivity": float(abs(sens)),
            "local_signed_sensitivity": float(sens),
            "step": float(step),
            "base_value": float(X_num_row[0, j]),
        })

    out = (
        pd.DataFrame(rows)
        .sort_values("local_abs_sensitivity", ascending=False)
        .head(n_top)
        .reset_index(drop=True)
    )
    return base, out


candidate_rows = []
for q in [
    analysis_train.query("case_type == 'TP'").sort_values("realmlp_
    analysis_train.query("case_type == 'TN'").sort_values("realmlp_
    analysis_train.query("case_type == 'FP'").sort_values("realmlp_
    analysis_train.query("case_type == 'FN'").sort_values("realmlp_
]:
    if len(q):
        candidate_rows.append(int(q.index[0]))

for ridx in candidate_rows:
    base_pred, local_sens = local_numeric_sensitivity_for_row_array
        row_index=ridx,
        realmlp_bundle=realmlp_bundle,
        fold_cache=fold_cache,
        features=[f for f in ["tenure", "MonthlyCharges", "TotalCha
                  if f in realmlp_bundle["NUMS"]],
        n_top=12,
        eval_bs=32768,
    )
    print({
        "row_index": ridx,
        "y_true": int(analysis_train.loc[ridx, "y_true"]),
        "pred": float(base_pred),
        "case_type": analysis_train.loc[ridx, "case_type"],
    })
    display(local_sens)
```

```
{'row_index': 213194, 'y_true': 1, 'pred': 0.9651018381118774, 'case
_type': 'TP'}
```

	feature	local_abs_sensitivity	local_signed_sensitivity	step
0	x3	2.148020e-02	-2.148020e-02	0.016756
1	x2	4.050195e-04	4.050195e-04	1.998573
2	x1_0	2.104229e-04	2.104229e-04	0.011330
3	x1_1	5.520270e-05	-5.520270e-05	1.255739
4	tenure	5.156148e-05	-5.156148e-05	1.253095
5	MonthlyCharges	2.183319e-05	2.183319e-05	1.553371
6	x4	3.045480e-07	-3.045480e-07	96.291824
7	TotalCharges	1.878855e-07	-1.878855e-07	117.695736

```
{'row_index': 284352, 'y_true': 0, 'pred': 0.0013824042398482561, 'c
ase_type': 'TN'}
```

	feature	local_abs_sensitivity	local_signed_sensitivity	step
0	x3	7.110491e-04	-7.110491e-04	0.016756
1	x1_0	1.442137e-04	-1.442137e-04	0.011330
2	x2	6.622617e-06	6.622617e-06	1.998573
3	x1_1	5.725560e-07	5.725560e-07	1.255739
4	tenure	4.450481e-07	4.450481e-07	1.253095
5	MonthlyCharges	3.327499e-07	-3.327499e-07	1.553371
6	x4	5.425983e-08	-5.425983e-08	96.291824
7	TotalCharges	3.971172e-08	-3.971172e-08	117.695736

```
{'row_index': 106705, 'y_true': 0, 'pred': 0.9608534574508667, 'case
_type': 'FP'}
```

	feature	local_abs_sensitivity	local_signed_sensitivity	step
0	x3	3.271386e-02	-3.271386e-02	0.016756
1	x2	4.718989e-04	4.718989e-04	1.998573
2	x1_0	2.262046e-04	2.262046e-04	0.011330
3	x1_1	6.467212e-05	-6.467212e-05	1.255739
4	tenure	6.240651e-05	-6.240651e-05	1.253095
5	MonthlyCharges	2.488370e-05	2.488370e-05	1.553371
6	x4	3.549965e-07	-3.549965e-07	96.291824
7	TotalCharges	2.271338e-07	-2.271338e-07	117.695736

```
{'row_index': 81118, 'y_true': 1, 'pred': 0.0022984042298048735, 'ca
se_type': 'FN'}
```

	feature	local_abs_sensitivity	local_signed_sensitivity	step
0	x3	1.687360e-03	-1.687360e-03	0.016756
1	x1_0	9.612771e-04	-9.612771e-04	0.011330
2	x2	8.716414e-06	8.716414e-06	1.998573
3	MonthlyCharges	3.982057e-06	3.982057e-06	1.553371
4	x1_1	1.440753e-06	1.440753e-06	1.255739
5	tenure	2.005759e-07	2.005759e-07	1.253095
6	TotalCharges	7.149267e-08	7.149267e-08	117.695736
7	x4	5.960294e-09	5.960294e-09	96.291824

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

12. Export tables

This section writes the central report artefacts for the RealMLP study.

In [34]:
```
outdir = Path(cfg.output_dir)

analysis_train.to_csv(outdir / "realmlp_analysis_train.csv", index=
analysis_test.to_csv(outdir / "realmlp_analysis_test.csv", index=Fa
replay["realmlp_fold_metrics_df"].to_csv(outdir / "realmlp_fold_met
replay["schema_df"].to_csv(outdir / "realmlp_schema_check.csv", ind
dec_realmlp.to_csv(outdir / "realmlp_deciles.csv", index=False)
perm_imp.to_csv(outdir / "realmlp_permutation_importance.csv", inde
sens_df.to_csv(outdir / "realmlp_finite_difference_sensitivity.csv"
ablation_df.to_csv(outdir / "realmlp_grouped_ablation.csv", index=F
uncertainty_summary.to_csv(outdir / "realmlp_uncertainty_summary.cs

summary = {
    "oof_auc": float(replay["realmlp_oof_auc"]),
    "oof_logloss": float(log_loss(bundle["y"], replay["realmlp_oof"
    "oof_brier": float(brier_score_loss(bundle["y"], replay["realml
    "n_numeric_features": int(schema["numeric_count"]),
    "n_categorical_features": int(schema["categorical_count"]),
    "top10_permutation_importance": perm_imp.head(10).set_index("fe
    "top10_finite_difference_sensitivity": sens_df.head(10).set_ind
}
(outdir / "realmlp_xai_summary.json").write_text(json.dumps(summary
summary
```

Out[34]:
```
{'oof_auc': 0.9171993209286087,
 'oof_logloss': 0.29869582337069117,
 'oof_brier': 0.0955834612250328,
 'n_numeric_features': 22,
 'n_categorical_features': 54,
 'top10_permutation_importance': {'x2': 0.0056478004188659225,
  'SeniorCitizen': 0.0013062768447774653,
  'x1_0': 0.00098563323021994,
  'x3': 0.0005300559637209101,
  'tenure_digit1': 0.00031854286668386855,
  'MonthlyCharges': 0.00020172727197842466,
  'TotalCharges': 0.00019811209607401015,
  'TotalCharges_digit2': 0.00018454198728912186,
  'MonthlyCharges_digit2': 6.568548744279035e-05,
  'TotalCharges_digit3': 5.456684662719091e-05},
 'top10_finite_difference_sensitivity': {'x1_0': 0.1098923310637474,
  'SeniorCitizen': 0.022135356441140175,
  'MonthlyCharges_digit2': 0.003959360998123884,
  'tenure_digit1': 0.0025660244282335043,
  'x2': 0.0012736959615722299,
  'TotalCharges_digit2': 0.0011614697286859155,
  'TotalCharges_digit3': 0.0006937597063370049,
  'TotalCharges_digit1': 0.000672676251269877,
  'MonthlyCharges_digit-1': 0.00042705165105871856,
  'TotalCharges_digit-1': 0.0003416793479118496}}
```

In [ ]: